Adarsh Sahu

Data Science | Analytics | Applied Machine Learning

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Summary

Production and Industrial Engineering undergraduate at NIT Jamshedpur applying machine learning and statistics to operations problems — forecasting, ranking and fraud detection — with an emphasis on designing an evaluation that can actually fail.

Education

National Institute of Technology Jamshedpur **2024 – 2028**
B.Tech, Production and Industrial Engineering *Jamshedpur, Jharkhand*
Coursework: Operations Research, Engineering Statistics, Production Management, Engineering Economics.

Experience

OA Helper | *Growth Intern* **Jul 2026 – Present**

Remote — Ranchi, Jharkhand

- Built the campus acquisition channel nationwide — student communities, coding clubs and campus representatives — to drive platform adoption across engineering colleges.
- Translated campus feedback into targeting and positioning decisions, working directly with the product team on user-acquisition strategy.

Projects

RailCross — Railway Crossing Delay Assistant *Python, scikit-learn, TypeScript, Next.js*

- Predicts railway level-crossing state from traffic and stopped-time signals, on an interactive map.
- Evaluated on a **crossing-level holdout** — 3 of 14 crossings withheld entirely from training — so figures describe *unseen* locations: ROC-AUC 0.888, F1 0.593 against 0.16 for a rule baseline.
- A calibrated `UNKNOWN` abstention band lifts accuracy on decided cases to ∼90%, declining rather than guessing on stale inputs. *Synthetic benchmarks only.*

PRAHARI — Digital Fraud Intelligence Platform *JavaScript, Naive Bayes, Rule Engine*

- Client-side scam detector with **two independent engines** — a weighted-signal rule engine covering 10+ fraud families in English, Hindi and Hinglish, and a character-trigram Naive Bayes classifier.
- Scored by **leave-one-out cross-validation** over a 48-message corpus whose 24 negatives are real bank OTP and KYC texts, with a threshold slider exposing the precision/recall trade-off. ET AI Hackathon 2026.

Search Ranking Model — Learning to Rank *Python, XGBoost Ranker, BM25, TF-IDF*

- XGBoost ranker trained on the MSLR-WEB10K benchmark (136 features, graded human relevance judgements), with BM25, TF-IDF, freshness and popularity features.
- Evaluated with **NDCG@10 and MRR rather than accuracy**, since position and graded relevance — not correctness — define a ranked list: NDCG@10 0.5581, MRR 0.8516.

Olist E-Commerce SQL Analytics *PostgreSQL, Supabase, Data Modelling*

- Modelled Brazil's Olist marketplace; queried with multi-table joins, CTEs, views and window functions.
- Found São Paulo the largest market by GMV and quantified the link between late delivery and review score.

Also

Projects: M5 Walmart demand forecasting (Prophet vs XGBoost) • IBM Telco churn prediction (0.84 ROC-AUC) • RAG study assistant (FAISS)

Competitions: XAVENIR 2026 — hallucination detection without pretrained models • MOVES 2026

Leadership: Core Team Member, Phocus NIT Jamshedpur • Volunteer, National Service Scheme

Technical Skills

Languages & Tools Python, SQL, JavaScript/TypeScript, pandas, scikit-learn, XGBoost, PostgreSQL, FastAPI, Git

Machine Learning & Statistics Classification, ranking (NDCG/MRR), anomaly detection, time-series forecasting, gradient boosting, calibration and abstention, cross-validation and leakage control, hypothesis testing, experiment design

AI Systems RAG, embeddings, FAISS **Strengthening** PyTorch, Docker, MLflow